Spectral line diagnostics – work split

M2 IRT, MC "Radiation" 2026–27

Goal. Tool that starts from a real spectrum, (1) fits and removes the continuum, (2) detects and identifies the lines, (3) measures them and derives the physical properties of the absorbing gas. **Check-ins:** Oct 14, Nov 18. **Oral:** Dec 9 (50% of grade, may be individual).

Scope (after check-in 1, Oct 14). *Absorption lines*, first in the SDSS spectrum of the A0 star SDSS J172637.24+264127.6 (3810–9210 Å, $R \approx 2000$, S/N ≈ 60). Detection keeps a sign option so emission spectra (planetary nebulae, suggested by the teachers) can be added later.

Pipeline and status.

Step	Module	Status
1. Read the spectrum	`io` (MAST query/download, SDSS reader)	done (Pablo)
2. Fit and remove the continuum	`continuum.fit_continuum`, `normalise`	done (Gaël)
3. Detect the significant dips	`detect.find_features`	**to do – Pablo**
4. Identify them (comb per species)	`linelist`, `detect.identify`	**to do – Gaël**
5. Equivalent widths	`ew.measure_ew`	done; continuum error to add
6. Curve of growth: N, b per species	`cog.fit_cog`	done (on synthetic data)

Identification by a comb. A shift multiplies all wavelengths by the same factor, $\lambda_{\rm obs} = \lambda_{\rm rest}(1+z)$, so the *ratios* of the wavelengths of one species are fixed. Each species is a comb of rest wavelengths; we slide it over the detected features (trial velocity v, -600 to $+600$ km/s for a star) and count the teeth that hit a feature. Hydrogen's comb comes from the Rydberg formula, $1/\lambda = R_H(1/n_{\rm low}^2 - 1/n_{\rm up}^2)$; other species from a table (NIST). Strength ratios reject chance matches: a strong line predicted but missing excludes the species; doublets (CaII H&K, NaD) must have EW ratios between 2:1 (weak) and 1:1 (saturated). Telluric O_2/H_2O bands are one more "species".

Pablo – data → features

- `detect.find_features`: significance of the absorption in a window of about one resolution element (or a matched Gaussian filter), threshold 4–5σ
- Merge neighbouring detections (a Balmer line with its wings = one feature); window, centre and its error, depth, EW
- Plot of the detected features on the spectrum
- Telluric band masks; later NIST queries for the line list
- Tests: synthetic spectra with known line positions (pytest)

Gaël – features → physics

- `linelist`: hydrogen comb from the Rydberg formula (`scipy.constants`); table of common species (CaII, MgII, NaI, OI, HeI, FeII) with $\lambda_{\rm vac}$, f, lower-level energy
- `detect.identify`: comb scan, scoring weighted by expected strength, strength-ratio checks
- Continuum error in `measure_ew`
- Curve of growth on the identified lines
- Tests: a hand-written features table of the star

Interface: the features table (full description in the docstring of the package `spectral_diag.detect`). `find_features(spectrum, threshold=4.0, sign=-1)` returns an astropy `Table`, one row per feature, columns `wave`, `wave_err`, `wave_min`, `wave_max` (Å, vacuum), `depth`, `ew`, `ew_err` (Å), `snr`. `identify` adds `species`, `line`, `wave_rest`, `v` (km/s). Each side can be written and tested without the other: `detect` is a package with one module per job, and `identify` goes in its own module `detect/identify.py`, re-exported by the package.

Timeline.

Until Oct 14	Repo, data classes, synthetic generator; profiles, EW, curve of growth validated on synthetic spectra.
Oct 14 – Nov 18	Real star: reading (done), continuum (done), detection (Pablo), line list and comb identification (Gaël), EWs with continuum error, curve of growth for CaII / Balmer. Demo of the full chain.
Nov 18 – Dec 8	Comb on a synthetic spectrum at unknown z; uncertainties; interactive viz; optional emission-line extension (planetary nebula); slides and rehearsal.
Dec 9	Oral presentation.

Working habits. Shared git repo, personal branches (`gael`, `pablo`) merged into `main` by reviewed pull requests; synthetic checks in `tests/` (run `pytest`), real-data notebooks in `notebooks/`; spectra in `data/` are not committed (re-download with `notebooks/MAST_query.ipynb`); weekly sync; review each other's code so both understand the full pipeline.